Calibrated Mixed Layer DO Product for 2018-2021

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This document provides an overview of the calibration process used to take uncalibrated, OOI-provided dissolved oxygen values from the near-surface instrument frame (NSIF) and surface mooring (SUMO) for the Irminger Sea and produce a calibrated, merged mixed-layer product for deployment years 5 through 8. The file GI_SUMO_Oxygen_Calibration.mlx provides the code to pull the data directly from the THREDDS server, process into 15 minute intervals, calibrate using turn-around cruises or calibrated co-located instruments, remove periods of biofouling, and create a merged mixed-layer DO product.

Assemble Data

Relevant HITL annotations for Deployments 5-8

- The use of UV lamps has been implemented for bio-fouling mitigation on this instrument as of the start date/time of this annotation. (April 2018)
- Note, no QARTOD tests are implemented at this time for oxygen data streams
- Several annotations about power outages and resulting data gaps
- CTD Deployment 5: Instrument unable to successfully power on and sample since 2/7/2019. Any data after this date are suspect. Instrument may be collecting data internally on battery power, but is unlikely.
- DO Deployment 7: Upon recovery it was noted that the power pin for the UV light cable was corroded. The light likely stopped being powered at some point during the deployment. This instrument did not have noticeable biofouling.
- CTD and DO Deployment 0008 - No Data Available

Download and initial processing

Download and initial processing of NSIF

- Data was pulled from THREDDS server on 8/2/2023
- Used Recovered files since telemetered didn't add any additional coverage
- Aanderaa sensor burst samples at 0.5 Hz for 3 minutes every 15 minutes at ~ 12 dbar depth
- The science idea behind 3 minute burst is it gives you enough data to average over a series of wave cycles, minimizing the impact of surface waves on the measurement- Chris Wingard, personal communication
- Additional parameters of interest were calculated using GSW Toolbox
- Data was formatted into a timetable with 15 minute intervals using a median value for the burst sampling

Download and initial processing of SUMO Buoy Data

- Data was pulled from THREDDS server on 8/15/2023
- Used both recovered and telemetered data for most complete coverage
- Aanderaa sensor samples at 2 sec intervals at ~1 dbar depth
- Data was formatted into a timetable with 15 minute intervals using a mean value for 15 minute interval

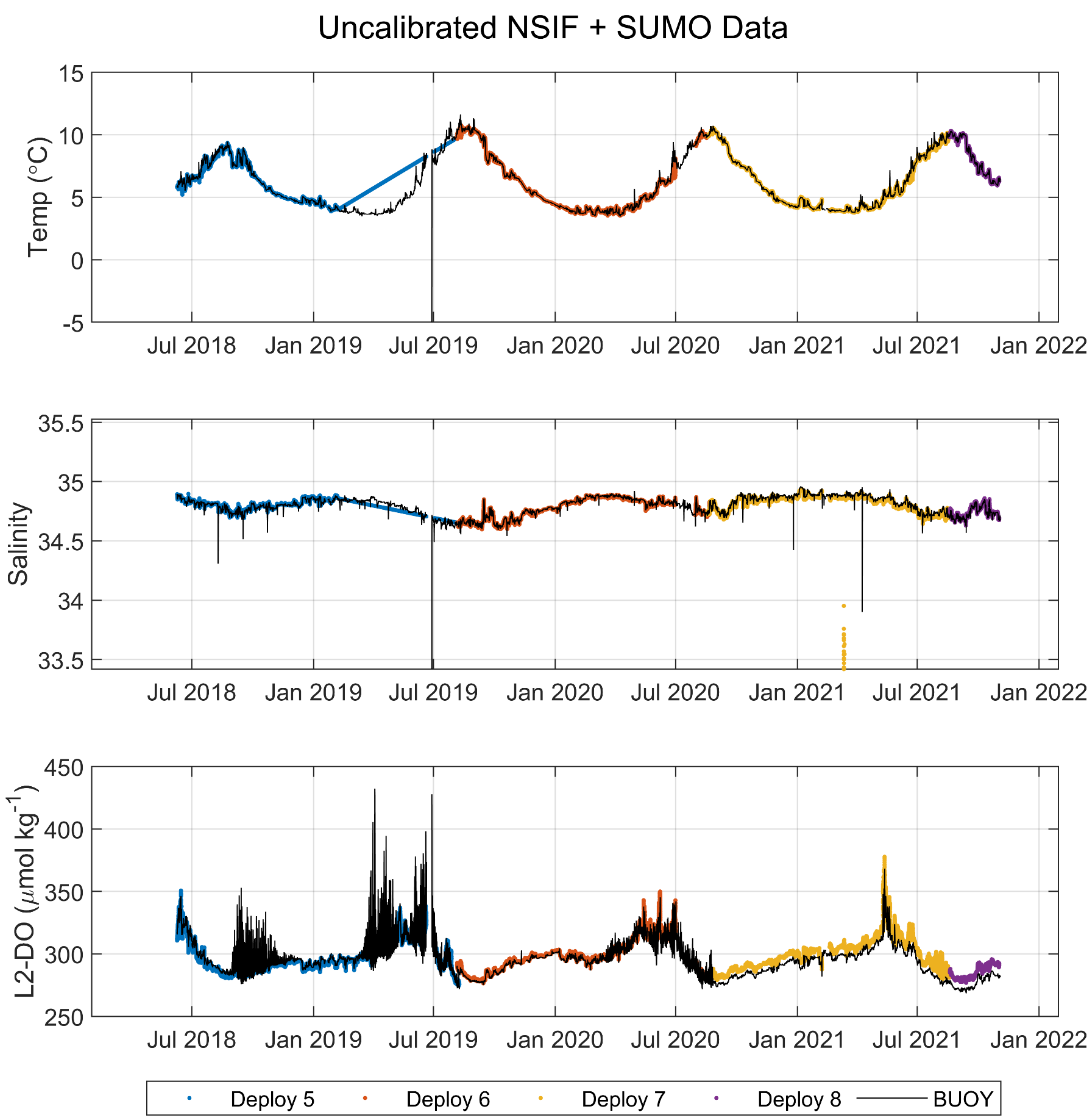

Figure 1: SUMO and NSIF values before calibration from co-located CTD and Aanderaa optode for NSIF (~12 dbar) in color and SUMO (~1 dbar) in black. Missing CTD temp and salinity data from Deployment 5 has been filled with dummy data in order to process oxygen data. Diurnal frequency in oxygen data indicative of biofouling on Aanderaa optode.

Temperature/Salinity corrections for oxygen processing

NSIF Deployment 5:

- Dummy temperature and salinity data for NSIF was replaced with temperature data from NSIF Aanderaa optode and salinity data from SUMO CTD
- L2- oxygen concentrations were recalculated with updated temp/sal data
- GSW parameters were recalculated
- Temperature difference of ~ 3 C results in ~1.8 umol/kg difference in recalculated Level 2 product

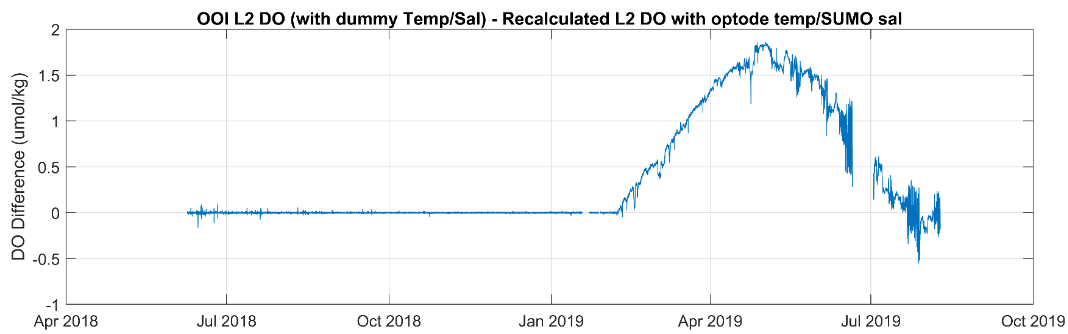

Figure 2: Difference between oxygen values calculated using dummy temp/salinity value and recalculated oxygen values using optode temperature and salinity for SUMO CTD for Deployment 5 NSIF Aanderaa optode.

Uncalibrated oxygen files

Data files processed to this point have been saved as mixed_layer_timetables_pregain.mat in G:\Shared drives\NSF_Irminger\OOI_DO_fixed_depth\Data\mixed_layer.

- The temp/sal corrections for deployment 5 have been saved in this file.
- Tables include additional parameters calculated using GSW Toolbox
- DOdt variables is the left edge averaged time window
- DOdn is the middle of the averaged time window
    - Middle of 15 minute window for SUMO
    - Middle of 3 minute burst sampling for NSIF
- Uncorrected L2 OOI-provided product (umol/kg) is 'dissolved_oxygen' variable

Calculate oxygen gain

When available, calibrated dissolved oxygen profiles from turn-around casts were used to determine Aanderaa storage drift and deployment drift. Casts within 10 km of OOI assets that overlapped in sampling time were identified. Asset measurements were matched with profile measurements using direct pressure and potential temperature values, as well as mean mixed layer values since both NSIF and SUMO assets are in the mixed layer. Mixed layer values were determined using both temperature and oxygen to avoid calibrating oxygen assets while located in the oxycline. Mixed layer depths were determined using CDT Toolbox.
Temperature-determined mixed-layer values were determined using Holte & Talley 2009

algorithm while oxygen-determined mixed-layer values were determined using the threshold method. The threshold method identified the bottom of the mixed layer as the first depth, measured from the reference depth down, where the profile value differs from the reference depth value by more than umol/kg.

During time periods when beginning or end of asset measurements did not overlap with CTD-DO profiles from turn-around cruises, gain calculations were made using co-located calibrated oxygen assets. For example, during Deployment Year 6, the data collected from the SUMO optode displayed signs of biofouling; therefore, no drift correction was made from dissolved oxygen profiles during the turn-around cruise. Instead, values from the storage and drift-corrected optode on the NSIF were used with collected data pre-biofouling signal for the SUMO optode to calculate a drift correction for the SUMO optode. This approach was also used to determine drift gain correction for the SUMO optode during Deployment 7 recovery (due to biofouling) and the storage gain correction for the SUMO optode during Deployment 8 deployment (no time overlapping turn-around casts). For optodes deployment during Deployment 8, no drift correction was done because both optodes on SUMO and NSIF stopped collecting data before recovery.

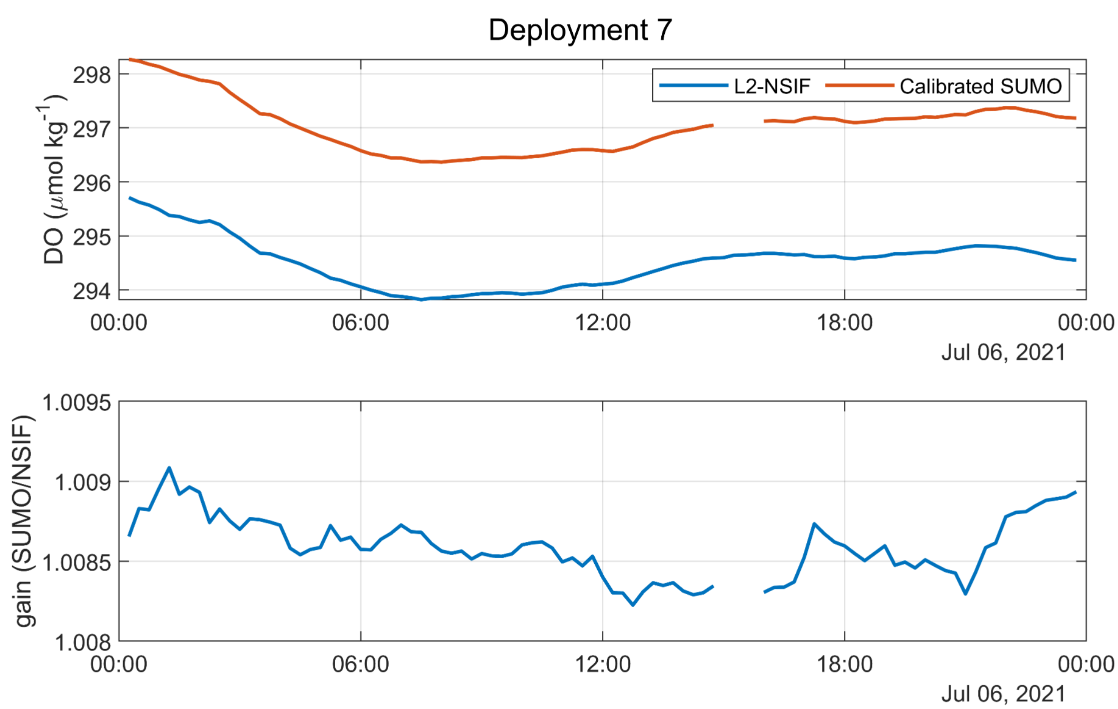

Figure 3: Example from Deployment 7 of using co-located and calibrated oxygen optode (top panel; red line; SUMO) to determine gain (bottom panel) for co-located and uncalibrated oxygen optode (top panel' blue line; NSIF).

Table 1: Summary of cast number or co-located oxygen asset used to determine gains for storage and deployment drift for each deployment year for NSIF and SUMO. Casts in red indicate that gain was calculated using an oxygen profile made with an Aanderaa optode. Cast number indicated under co-located asset indicates the cast used to calibrate that co-located asset.

Deploy. Year	NSIF (~12 dbar)		SUMO (~1 dbar)		Notes
	Storage Gain	Drift Gain	Storage Gain	Drift Gain	
5	Cast 5	Cast 3	Cast 5	Cast 3	Cast 3 = AA optode profile
6	Cast 11	Cast 9	Cast 11	NSIF (Cast 9)	Cast 11 = AA optode profile Cast 9 = AA optode profile
7	Cast 9	SUMO (Cast 5)	Cast 9	Cast 5	Cast 9 = AA optode profile
8	SUMO (Cast 5)	--	Cast 5	--	No drift correction done

Mixed-layer DO product

- Gain calculations were applied to correct for storage drift and deployment drift
- Periods of observed biofouling were removed for each sensor
- The mean of remaining NSIF and SUMO calculated for each deployments
- Deployments were concatenated and the mean of temporally overlapping values was calculated
- Result is time series with resolution of 15 minutes for dissolved oxygen in the mixed layer for deployments years 5 to 8
- This product is saved as **mixed_layer_calibrated_oxygen.mat** in G:\Shared drives\NSF_Irminger\OOI_DO_fixed_depth\Data\mixed_layer
- Tables have been saved as:
    - Asset and deployment year (ex. buoy_DO5 or nsif_DO5)
        - Includes lower levels OOI-provided product

- A merged mixed layer table for each deployment year (ex. ML_DO5)
    - Doesn’t include lower level OOI products or uncalibrated oxygen
- A merged mixed layer table for all deployment years (ML_DO)
    - Doesn’t include lower level OOI products, uncalibrated oxygen, or gain values

- Key variables in all:
  - ‘DOdt’ = datetime of left edge of averaged time window (ex. 13:00 is mean for 13:00 - 13:15 window)
  - ‘DOdn’ = mean date number for the time window (ex. 13:07:30 is the mean for the 13:00 -13:15 window)
  - ‘dissolved_oxygen’ = uncalibrated L2 OOI provided oxygen product (umol/kg)
  - T, SP, SA, CT, pt, O2sol_umolkg, rho, prho, sigma0 have been calculated using GSW Toolbox
- In tables by asset and year:
  - ‘gain_ts’ = the calculated gain value for each time interval
  - ‘DOcorr_umolkg’ = gain corrected oxygen value (dissolved_oxygen * gain_ts)
  - ‘DOqaqc_umolkg’ = gain corrected oxygen value with biofouling periods removed
- In mixed layer table:
  - ‘DO_umolkg_final’ = final oxygen value (gain corrected and biofouling removed)

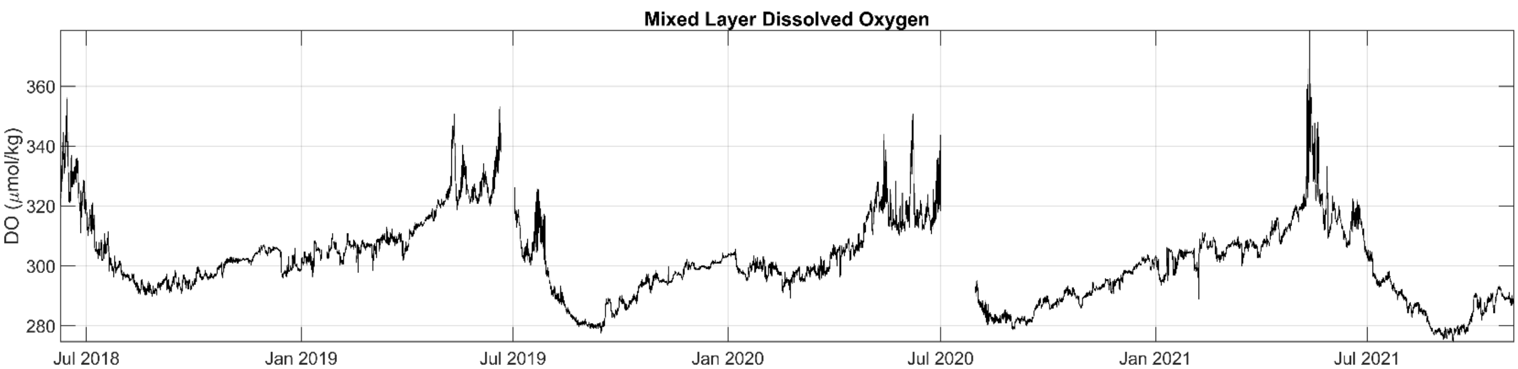

Figure 4: Finalized, calibrated mixed-layer product.

References

Greene, Chad A., Kaustubh Thirumalai, Kelly A. Kearney, José Miguel Delgado, Wolfgang Schwanghart, Natalie S. Wolfenbarger, Kristen M. Thyng, David E. Gwyther, Alex S. Gardner, and Donald D. Blankenship (2019). The Climate Data Toolbox for MATLAB. Geochemistry, Geophysics, Geosystems, 20, 3774-3781. [doi:10.1029/2019GC008392](https://doi.org/10.1029/2019GC008392)

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